

MANAVA Multiverse Discord Bot

QA Report & Acceptance Test Handover

✓ 89 / 89 automated tests passing · type-check clean

PROJECT	MANAVA Multiverse — custom Discord bot (squads, XP economy, leaderboards, application workflows, MANAVA Gateway integration)
DOCUMENT	QA Report & Acceptance Test Handover (Milestone 3 deliverable)
BUILD / COMMIT	97a8b39 (main)
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RUNTIME	Python 3.11+ · discord.py 2.7.1 · Supabase Postgres (asyncpg)
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RECIPIENT	MANAVA

This document accompanies the source repository and the docs/ handover set (command-reference.md , schema.md , deployment.md , recovery.md , manava-gateway-integration.md , logging.md , known-limitations.md , deferred-items.md). It supersedes the plain-text docs/qa-report.md for presentation purposes; the two carry the same substance.

1. Document purpose & contents

This report is the QA deliverable for Milestone 3. It states what has been verified automatically, provides the full evidence inventory, and hands MANAVA an executable acceptance-test checklist for the items that require a live Discord server and the real MANAVA Gateway credentials.

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2. System under test

A single-process Discord bot (discord.py 2.x, fully async) backed by a Supabase Postgres database accessed directly via `asyncpg`. It provides:

- **Squads** — eligibility-gated creation, per-squad role + four channels distributed across the `SQUADS 01-07` categories, membership and leadership flows (apply / invite / approve / reject / leave / remove / promote / demote / transfer / disband), 30-day archive on disband, and `/admin restore-squad` recovery.
- **Unified XP economy** — one Personal XP balance (Discord text activity + MANAVA events), Lifetime + Season counters, 1:1 cascade into Squad XP, level-based member caps.
- **Leaderboards** — global squad (Season XP, top-32 badge) and per-squad member, rendered natively.
- **Applications** — Developer and Creator/Streamer review workflows with persistent Approve / Reject / Request-More-Info controls and a persistent applicant reply flow.
- **Seasons** — manual Start / End / Start-New (resets Season XP, never Lifetime).
- **MANAVA Gateway integration** — identity lookup (`GET /internal/identity/:discordId`) and HMAC-SHA256-signed inbound event webhooks, verified against MANAVA's Gateway source.

Two operating modes

Gateway mode (when `MANAVA_GATEWAY_BASE_URL` + `DISCORD_BACKEND_API_KEY` are set) uses the real MANAVA identity service and HMAC webhook verification. **Stub mode** (default until MANAVA supplies credentials) uses the manual `/admin set-verified` and `/admin link-manava-account` overrides and a legacy bearer-token webhook. The automated suite covers both; the live checklist in §6 is written for stub mode with a Gateway-mode addendum.

3. Test approach & how to run

3.1 Automated (this report, §4–§5)

`pytest` with `pytest-asyncio`. Tests are unit / integration level: service logic, permission-overwrite construction, event parsing and de-duplication, HMAC signature verification, cross-squad isolation, disband cleanup and recovery, application flows, season reset, and the protected-role guard. Repositories and Discord objects are mocked, so the suite needs no live database or gateway connection and runs in ~1 second. `mypy` is run as a type-safety gate.

Step	Command
Install	<code>python -m pip install -r requirements-dev.txt</code>
Run tests	<code>pytest -q</code>
Type check	<code>mypy src</code>

Both also run automatically as the `quality` stage in `.gitlab-ci.yml`.

3.2 Manual / live-server (§6)

Items that depend on a real Discord guild, real Discord API side effects (role/channel creation, DMs), a running database, or the real MANAVA Gateway are listed as an executable checklist in §6 with explicit steps, expected results, and a result column for the tester to complete.

4. Automated test results

89 TESTS	89 PASSING	0 FAILING	0 MYPY ERRORS
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89 passed in 1.34s · mypy src : Success: no issues found in 68 source files.

4.1 Results by test file

Test file	Tests	Area covered	Result
test_eligibility.py	12	Squad-name validation; linked/verified failure message includes the MANAVA link URL when configured	PASS
test_capacity.py	4	SQUADS category assignment: first with room, never by squad count, raise when all full	PASS
test_membership_flows.py	4	Officer limit (max 2); officers can't promote peers; leader can't leave without transfer/disband; leader can't be removed	PASS
test_permission_isolation.py	3	Cross-squad isolation — officer overwrites are per-member and never leak between squads; member channels key off the squad-specific role only	PASS
test_disband_cleanup.py	1	Disband strips the shared Squad Leader/Officer/Member rank roles from every member	PASS
test_squad_restore.py	5	/admin restore-squad — happy path, rejects active / purged / leader-elsewhere, rolls back the recreated role on DB failure	PASS
test_squad_xp_contribution.py	4	Squad level advances past multiple thresholds / stays at L1 below the first; member-capacity allows below cap, blocks when full	PASS
test_discord_xp_cooldown.py	8	Text-XP eligibility: 60s cooldown (before/at boundary/after), channel & category exclusion, unrelated exclusions don't block	PASS
test_manava_event_dedup.py	2	Duplicate event_id never grants XP twice ; an unresolved MANAVA user grants no XP	PASS
test_manava_event_parsing.py	5	Gateway camelCase payloads (match / placement + wonPrizeSlot); legacy snake_case & renamed types accepted; missing field / unknown type rejected	PASS

Test file	Tests	Area covered	Result
test_gateway_signature.py	7	HMAC x-manava-signature : valid passes, tampered/wrong-secret/missing fails, signing secret beats legacy bearer, byte-for-byte match with the Gateway's hmac-signer.ts	PASS
test_gateway_identity.py	8	Identity payload mapping; link_state = OK / NOT_LINKED / NOT_VERIFIED; caches to personal_xp ; falls back to cached value on Gateway error; stub-mode path	PASS
test_leaderboard_queries.py	2	Global & member leaderboard row→entry mapping (single query, no N+1)	PASS
test_season_reset.py	3	Start-New zeroes Personal + Squad Season XP only (Lifetime untouched), actor audited; no-active-season path; end-with-none raises	PASS
test_application_flows.py	6	Submit writes history + audit with no PII in the audit row ; approve is one-way; more-info round-trip; role mapping (Developer / Creator / Streamer / Both); bad email creates nothing	PASS
test_application_cooldown.py	4	One active application per type; 30-day post-reject cooldown enforced; past-cooldown and no-prior-reject allow	PASS
test_application_permissions.py	3	Reviewer boundary — Admin/MANAVA Team elevated; Moderator/Senior Moderator not; server-Administrator bootstrap fallback	PASS
test_application_channel_config.py	2	Per-type review-channel override round-trip; finalize_submission posts to the configured channel and falls back to #applications-review	PASS
test_protected_role_guard.py	6	Least privilege — the bot refuses to grant Admin / MANAVA Team / Senior Moderator / Moderator, before any Discord call; non-protected roles allowed	PASS

4.2 Acceptance-criteria coverage matrix

Acceptance criterion (project brief)	Automated coverage	Live check
Eligible squad creation; every individual ineligible gate	name & link-message gates (test_eligibility)	\$6.1
Duplicate name rejection; second-squad rejection	name reservation; one-active-membership index paths (test_membership_flows)	\$6.1
Apply / Invite / Approve / Reject	approve/reject re-check & officer-limit logic	\$6.2
Leave / Remove; Promote / Demote; Transfer Leadership	test_membership_flows (guards); leadership never lost	\$6.2
Disband / archive	test_disband_cleanup ; archive model doc	\$6.4

Acceptance criterion (project brief)	Automated coverage	Live check
Category capacity enforcement	test_capacity (4)	—
Permission isolation between two squads	test_permission_isolation (3)	\$6.3
Valid text grants XP; too-short doesn't; cooldown enforced	test_discord_xp_cooldown (8)	\$6.5
MANAVA event grants XP; duplicate never grants twice	test_manava_event_dedup , test_manava_event_parsing	\$6.6 / \$6.10
Member vs non-member contribution; leave-then-join behaviour	test_squad_xp_contribution ; xp_service cascade design	\$6.5
Wave 1500 gate	gate wiring (squad_eligibility)	\$6.1
Season start / end / reset; Lifetime persists	test_season_reset (3)	\$6.7
Application submission; Developer Pending assignment	test_application_flows	\$6.8
Request-More-Info round-trip; Approve / Reject; role assignment	test_application_flows (round-trip, one-way, role map)	\$6.8
30-day reapplication cooldown	test_application_cooldown (4)	\$6.8
Reviewer-permission boundary (Mod / Senior Mod blocked)	test_application_permissions (3)	\$6.9
Application PII never in generic logs	test_application_flows asserts audit row is PII-free	—
Bot never grants a protected role	test_protected_role_guard (6)	—
Inbound webhook authentication	test_gateway_signature (7)	\$6.10

5. Automated test inventory (evidence)

Every test function, grouped by file. All 89 pass.

Squads — eligibility, capacity, membership, isolation, lifecycle

Test	Asserts
<code>test_validate_squad_name</code> ×10	Length 3–32 and character-set rules accept/reject the right names
<code>test_link_failure_message_includes_url_when_configured</code>	“link your account” / “confirm your email” text carries <code>MANAVA_LINK_URL</code> when set, generic otherwise
<code>test_picks_first_category_with_room</code>	Assignment fills categories in order, purely by capacity
<code>test_never_assigns_by_squad_count_beyond_capacity</code>	10-squad / 40-channel cap per category is respected
<code>test_empty_counts_defaults_to_first_category</code>	First squad lands in <code>SQUADS 01</code>
<code>test_raises_when_all_categories_full</code>	70-active-squad ceiling raises a friendly capacity error
<code>test_promote_blocked_at_officer_limit</code>	A 3rd officer promotion is rejected (max 2)
<code>test_officer_cannot_promote_peers</code>	Only the Leader can promote
<code>test_leader_cannot_leave_without_transfer_or_disband</code>	Leader-leave is blocked — a squad can never end up leaderless
<code>test_cannot_remove_the_leader</code>	The Leader cannot be removed by an officer
<code>test_officer_overwrites_for_one_squad_never_include_another_squads_leader</code>	Squad A's officer overwrites contain only Squad A people
<code>test_officer_overwrites_never_leak_between_squads_officer_sets</code>	Two squads' officer sets are disjoint in the built overwrites

Test	Asserts
<code>test_member_overwrites_key_off_squad_specific_role_only</code>	Member channels are gated by the squad-unique role, not the shared one
<code>test_disband_strips_shared_rank_roles_from_every_member</code>	On disband each member loses Squad Leader/Officer/Member; the squad role is deleted
<code>test_restore_happy_path</code>	Restore recreates the role, marks active, re-grants access, audits
<code>test_restore_rejects_active_squad / ..._after_channels_purged / ..._when_former_leader_is_in_another_squad</code>	Restore refuses the three invalid states with a user-facing message, no side effects
<code>test_restore_rolls_back_recreated_role_if_db_step_fails</code>	A DB failure after role creation deletes the orphan role (no partial state)
<code>test_squad_level_advances_past_multiple_thresholds / ..._stays_at_1_below_first_threshold</code>	Level derived correctly from Lifetime Squad XP vs configurable thresholds
<code>test_capacity_allows_new_member_below_cap / ..._blocks_new_member_when_full</code>	Per-level member cap enforced at join time

XP economy & leaderboards

Test	Asserts
<code>test_no_prior_grant_is_never_on_cooldown</code>	First message from a user is always eligible
<code>test_blocked_within_cooldown_window / test_allowed_exactly_at_boundary / test_allowed_once_cooldown_elapses</code>	60-second per-user cooldown boundary behaviour
<code>test_channel_directly_excluded / test_category_excluded_blocks_channel_inside_it</code>	Channel and parent-category exclusion both suppress XP
<code>test_no_category_and_not_excluded_is_eligible / test_unrelated_exclusions_do_not_block</code>	Only listed channels/categories are excluded
<code>test_duplicate_event_id_never_grants_xp_twice</code>	Re-delivering the same <code>event_id</code> results in exactly one XP grant — hard correctness requirement

Test	Asserts
test_unresolved_manava_user_grants_no_xp	An event for an unknown MANAVA user is recorded but grants nothing
test_parses_gateway_match_completed_payload / ... _placement_payload_with_prize_slot	Gateway camelCase payloads parse, incl. place and wonPrizeSlot
test_legacy_snake_case_and_renamed_types_still_accepted	Pre-Gateway payloads and old event-type names still work (aliases)
test_missing_required_field_is_rejected / test_unknown_event_type_is_rejected	Malformed payloads produce a validation error, not a crash
test_global_leaderboard_maps_rows_to_entries / test_member_leaderboard_maps_rows_to_entries	Leaderboard queries map to typed entries in one pass

MANAVA Gateway integration

Test	Asserts
test_valid_signature_passes / test_tampered_body_fails / test_wrong_secret_fails / test_missing_signature_header_fails	HMAC-SHA256 x-manava-signature verification accepts only an untampered body signed with the shared secret
test_matches_gateway_signpayload_wire_format	Our verification is byte-identical to the Gateway's hmac-signer.ts over a realistic payload
test_legacy_bearer_used_only_without_signing_secret / test_signing_secret_takes_precedence_over_legacy	Clean cut-over: HMAC when the signing secret is set, legacy bearer only as a fallback
test_identity_from_payload_linked / ... _not_linked_blanks_other_fields	Identity response mapping, including the not-linked case
test_link_state_distinguishes_unverified_from_unlinked / ... _reports_not_linked_from_gateway / ... _reports_not_verified_from_gateway	link_state returns OK / NOT_LINKED / NOT_VERIFIED so the squad gate shows the right guidance
test_is_verified_uses_gateway_and_caches	A successful lookup caches manava_user_id + verified_player into personal_xp
test_is_verified_falls_back_to_cache_on_gateway_error / test_is_verified_stub_mode_when_no_gateway	Gateway outage or stub mode falls back to the cached / manual value

Applications & seasons

Test	Asserts
<code>test_submit_records_history_and_keeps_pii_out_of_audit</code>	Submission writes an <code>application_history</code> row and an audit row; the audit row contains no email / free-text PII
<code>test_submit_rejects_bad_email_and_creates_nothing</code>	An invalid contact email (or URL) is rejected server-side; <code>applications.insert</code> is never called
<code>test_submit_blocked_by_existing_active</code>	A second active application of the same type is refused
<code>test_approve_is_one_way</code>	A decided application cannot be re-decided
<code>test_more_info_round_trip</code>	Request-More-Info → applicant reply attaches to the same record and returns it to pending; only the real applicant can reply
<code>test_roles_for_approval_and_pending_role</code>	Developer → Developer (+ Pending); Creator / Streamer / Both → the right role(s), no Pending
<code>test_active_application_blocks_new_one /</code> <code>test_recent_rejection_is_in_cooldown /</code> <code>test_rejection_past_cooldown_allows_reapply /</code> <code>test_no_prior_rejection_allows</code>	30-day post-reject cooldown, enforced server-side
<code>test_moderator_and_senior_moderator_are_not_elevated /</code> <code>test_admin_and_manava_team_are_elevated /</code> <code>test_server_administrator_permission_is_a_bootstrap_fallback</code>	Only Admin / MANAVA Team (or a server Administrator) pass the review gate
<code>test_set_and_get_review_channel_roundtrip</code>	Per-type review-channel override stores and clears correctly in <code>bot_config</code>
<code>test_finalize_uses_configured_channel_then_falls_back</code>	A configured type posts to its channel; an unconfigured type uses <code>#applications-review</code>
<code>test_start_new_season_resets_season_xp_only</code>	Start-New zeroes Personal + Squad Season XP; Lifetime values are untouched; actor recorded
<code>test_start_new_season_with_no_active_season</code>	Start-New works even with no season currently active

Test	Asserts
<code>test_end_season_with_none_active_raises</code>	Ending with no active season raises a user-facing error

Security

Test	Asserts
<code>test_ensure_assignable_blocks_every_protected_role</code> ×4	Admin, MANAVA Team, Senior Moderator, Moderator each raise on an attempted grant
<code>test_ensure_assignable_allows_non_protected_roles</code>	Squad / application roles are grantable
<code>test_assign_role_refuses_protected_before_touching_discord</code>	The deny-list check happens before any Discord API call — no partial side effect

6. Manual / live-server acceptance checklist

Run once against a staging guild with the bot deployed and migrations applied. Unless noted, these use **stub mode** (Gateway env vars unset) so `/admin set-verified` acts as the linked/verified gate. You will need 2–3 Discord accounts (or alts) for the membership flows. Tick *Result* as Pass / Fail and add notes.

#	Case	Steps	Expected	Result
6.1	Squad creation gate — each failure	With XP < 1500 / no Member role / unverified / already in a squad / a duplicate name, run <code>/squad create</code> .	Each failed gate is listed in one plain-language message; no squad created.	
6.2	Eligible creation + membership flow	<code>/admin set-verified</code> + <code>/admin add-xp 1500</code> , then <code>/squad create</code> . Second account <code>/squad apply</code> → approve via the DM buttons. Then <code>/squad invite</code> , <code>/squad promote</code> (3rd → blocked), <code>/squad demote</code> , <code>/squad remove</code> , <code>/squad transfer-leadership</code> .	Role + 4 channels created under a SQUADS category; applicant DM'd on approve; officer cap of 2 enforced; every destructive action asks to confirm; leadership always present.	
6.3	Cross-squad isolation	Create two squads with different accounts. As an officer of squad A, try to open squad B's Officer Text channel.	Not visible / no access.	
6.4	Disband & archive; recovery	<code>/squad disband</code> (confirm). Then <code>/admin restore-squad squad_name:...</code> .	Access revoked immediately, squad off <code>/leaderboard squads</code> , rank roles removed; channels remain (30-day window). Restore re-activates it, re-adds members, re-grants access.	
6.5	Discord text XP	Post a ≥10-char message in a normal channel; repeat within 60s; post a <10-char message; post in an excluded channel.	+15 XP once; no XP on the repeat, the short message, or the excluded channel. Squad XP rises 1:1 while you're in a squad.	

#	Case	Steps	Expected	Result
6.6	MANAVA event pipeline (harness)	<code>/admin link-manava-account</code> , then <code>/xpconfig simulate-manava-event event_type:match_completed</code> . Repeat with a fixed <code>event_id</code> twice. Try <code>tournament_placement place:1</code> .	First run: status=processed , XP granted, Personal + Squad XP rise. Second identical run: status=duplicate , xp_granted=0 . Placement grants participation + reward.	
6.7	Seasons	Earn some XP. <code>/season start</code> , then <code>/season new</code> (confirm), then <code>/season status</code> .	Personal Season XP and Season Squad XP → 0 for all; Lifetime unchanged; an <code>audit_logs</code> row with <code>action = season.start_new</code> and the actor.	
6.8	Developer application, end to end	<code>/apply developer</code> → fill the form (try a bad email → Fix & resubmit) → submit. In <code>#applications-review</code> : Request More Info → applicant replies via the DM button → Approve . Then <code>/apply developer</code> again immediately; and again after a Reject.	Developer Pending assigned; embed posted; a bad email blocks with Fix & resubmit; more-info reply attaches to the same record; on approve Pending→Developer and applicant DM'd; a 2nd active application is blocked; after a Reject the 30-day cooldown message appears.	
6.9	Reviewer permission boundary	As an account holding only <i>Moderator</i> (or Senior Moderator), click a review button; also check <code>/admin</code> visibility.	Review action refused ("Only Admin / MANAVA Team..."). <code>/admin</code> , <code>/xpconfig</code> , <code>/season</code> are hidden in the command picker for non-Administrators.	

#	Case	Steps	Expected	Result
6.10	Persistence across restart	Submit a squad application and an app-review item, then restart the bot , then use the previously-posted buttons.	The Approve / Reject / Request-More-Info / Respond buttons still work after the restart.	
6.11	DM-closed handling	With server DMs disabled, run <code>/squad apply</code> ; and be the target of a squad-join or application decision.	<code>/squad apply</code> is blocked with instructions to enable DMs; a decision maker is told the applicant couldn't be DM'd.	
6.12	<i>Gateway mode</i> (once MANAVA supplies credentials + registers the URL)	Set <code>MANAVA_GATEWAY_BASE_URL</code> / <code>DISCORD_BACKEND_API_KEY</code> / <code>WEBHOOK_SIGNING_SECRET</code> , restart. <code>/admin gateway refresh-identity user:@you</code> . Have MANAVA fire a real event; then have it redelivered.	<code>/admin status</code> shows "Gateway" mode + HMAC auth. Identity returns the expected linked/verified. The real event grants XP and writes a <code>processed_events</code> row; the redelivery grants nothing.	

7. Failure-scenario coverage

Scenario	Handling	Evidence
Duplicate / retried MANAVA event	<code>event_id</code> reserved in <code>processed_events</code> before any XP is computed — guaranteed single grant	<code>test_manava_event_dedup</code> ; §6.6
Malformed / unauthenticated webhook	Bad HMAC → 401; malformed body → 400 (Gateway retries then dead-letters)	<code>test_gateway_signature</code> , <code>test_manava_event_parsing</code>
MANAVA / Gateway unavailable	Identity lookup falls back to the cached / manual value; the Gateway's own outbox re-delivers events when it recovers	<code>test_gateway_identity</code>
Supabase outage	Commands surface a friendly error; the webhook returns 503 so the Gateway retries; <code>with_db_retry</code> covers transient blips; <code>/healthz</code> stays up	<code>src/db/retry.py</code> , <code>src/web/webhook_server.py</code>
Bot restart mid-flow	Persistent views re-registered in <code>setup_hook</code> ; button state is carried in the <code>custom_id</code> / re-resolved from the DB, not held in memory	§6.10
Missing Discord permission / deleted channel or role	<code>role_safety</code> maps <code>Forbidden</code> to a friendly message; a deleted review channel falls back to <code>#applications-review</code> ; a deleted squad channel raises a “contact staff” error rather than crashing	<code>test_protected_role_guard</code> ; recovery doc
Partial squad create / delete	Discord objects are rolled back if the DB insert fails; restore deletes the recreated role if its DB step fails	<code>test_squad_restore</code> ; <code>squad_lifecycle.create_squad</code>
Lost leadership	Leader-leave blocked; transfer always assigns exactly one Leader; disband closes all memberships	<code>test_membership_flows</code>
Duplicate bot instance	Documented — single instance only, no leader election	§8

8. Known limitations & items pending external input

8.1 Accepted / by design

Item	Detail
Discord text-XP cooldown is in-memory	The 60-second per-user cooldown resets on restart — at most one extra grant right after a redeploy. Accepted trade-off.
Text-XP grants are best-effort during a DB outage	Logged and dropped, not queued.
Application field validation is format-only, post-submit	Discord modals can't validate inline; email / URL are checked on submit, with a "Fix & resubmit" recovery. No deliverability checks — reviewer's call.
Application modal is a 3-step flow	Discord's 5-input modal limit forces select menus → modal → optional comments.
Season resets are not snapshotted	The reset is audited (actor + counts) but final standings aren't stored. Export the leaderboard before <code>/season new</code> if a record is wanted.
<code>contributed_xp</code> is lifetime-of-stint, not seasonal	Per spec, a season reset does not touch it.
Archived squads keep inaccessible channels for 30 days	Then a daily job purges them. Recovery via <code>/admin restore-squad</code> within the window.
Single instance only	Two processes on the same token would double-process. No leader election.
Gateway identity cache is lazy	An event for a MANAVA user the bot has never looked up is recorded as <code>unresolved_user</code> and grants no XP until an identity lookup warms the cache. Client confirmed acceptable for launch.

8.2 Pending external input (not verifiable until MANAVA provides it)

Item	Owner
Real <code>DISCORD_BACKEND_API_KEY</code> and <code>WEBHOOK_SIGNING_SECRET</code>	MANAVA
Registering this deployment's <code>/webhooks/manava</code> URL with the Gateway (guarded by MANAVA's internal key)	MANAVA
A value for <code>prize_slot_bonus_xp</code> (currently <code>0</code> — <code>wonPrizeSlot</code> is inert until set via <code>/xpconfig set-xp</code>)	MANAVA
Production MANAVA account-linking URL for <code>MANAVA_LINK_URL</code> (demo URL is configured)	MANAVA
Live end-to-end run of the \$6 checklist on the staging guild	MANAVA QA

9. Sign-off

The automated suite (89 tests) and type check pass on build `97a8b39` . The manual acceptance checklist in §6 is ready for execution on the staging guild.

Role	Name	Signature	Date
Developer (build & automated QA)	Muhammad Nadeem		
MANAVA QA (§6 execution)			
MANAVA acceptance			